

Rajalakshmi Engineering College

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Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q2

Attempt : 1

Total Mark : 10

Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Elsa, a busy professional, is using a scheduling application to plan her meetings efficiently. The application requires users to input meeting durations in minutes, ensuring that the duration is a positive integer and does not exceed 240 minutes (4 hours). Elsa needs a program to assist her in scheduling meetings securely with proper exception handling.

Create a Java class named `ElsaMeetingScheduler`. Implement a custom exception: `InvalidDurationException` for invalid meeting duration entries. Implement the main method to interactively take user input for a meeting duration. Implement the `validateMeetingDuration` method to validate the meeting duration based on the specified rules and throw a custom exception if the validation fails. Print appropriate success or error messages based on the meeting duration.

Implement a custom exception, `InvalidDurationException`, to handle cases where the entered meeting duration does not meet the specified criteria.

Input Format

The input consists of an integer value 'n', representing the meeting duration.

Output Format

The output is displayed in the following format:

If the entered meeting duration meets the specified criteria, the program outputs

"Meeting scheduled successfully!"

If the entered meeting duration is invalid, the program outputs an error message indicating the issue.

"Error: Invalid meeting duration. Please enter a positive integer not exceeding 240 minutes (4 hours)."

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 120

Output: Meeting scheduled successfully!

Answer

```
import java.util.Scanner;
```

```
class InvalidDurationException extends Exception {  
    public InvalidDurationException(String message) {  
        super(message);  
    }  
}
```

```
class MeetingValidator {  
    public void validateMeetingDuration(int duration) throws  
        InvalidDurationException {  
        if (duration <= 0 || duration > 240) {
```

```
        throw new InvalidDurationException(
            "Invalid meeting duration. Please enter a positive integer not exceeding
            240 minutes (4 hours).");
    }
}
```

```
class MeetingScheduler {
    private MeetingValidator validator = new MeetingValidator();

    public void scheduleMeeting() {
        Scanner scanner = new Scanner(System.in);

        try {
            int meetingDuration = scanner.nextInt();

            validator.validateMeetingDuration(meetingDuration);

            System.out.println("Meeting scheduled successfully!");
        } catch (InvalidDurationException | java.util.InputMismatchException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}

public class Main {
    public static void main(String[] args) {
        MeetingScheduler scheduler = new MeetingScheduler();
        scheduler.scheduleMeeting();
    }
}
```

Status : Correct

Marks : 10/10